

Ethan Massey

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EDUCATION

Massachusetts Institute of Technology

Bachelor of Science in Mathematics and Computer Science

GPA: 4.8/5.0

Relevant Coursework: Design and Analysis of Algorithms, Introduction to Algorithms, Fundamentals of Programming, Mathematics for Computer Science, Probability and Random Variables, Real Analysis

Cambridge, MA

Expected May 2028

Dallas College

Associate of Science

GPA: 4.0/4.0; 80 Credit Hours

Dallas, TX

May 2025

TECHNICAL SKILLS

Programming Languages: Python, Java

Software Development: Git, Flask, Pygame, Pygbag, Folium; web applications and browser deployment

AI & Data: AI literacy fundamentals, prompt engineering, machine learning, data visualization, Bayesian modeling

Libraries & Tools: NumPy, pandas, scikit-learn, XGBoost, Optuna, SHAP

Certification: Information Technology Specialist: Java, May 2024

EXPERIENCE & PROJECTS

MIT Senseable City Lab | Undergraduate Researcher

Aug 2025 - Sep 2026

Research Repository

- Built an end-to-end **Python data pipeline** to ingest, process, and integrate 15 years of epidemiological, climate, demographic, and geospatial data across **92 municipalities** for spatiotemporal forecasting.
- Developed an interactive **Flask/Folium web application** to visualize municipality-level forecasts, historical outbreaks, and spatial disease patterns.
- Implemented temporal-lag and graph-based spatial feature pipelines using municipality adjacency networks; automated model training and hyperparameter optimization with **XGBoost and Optuna**.
- Implemented hierarchical Bayesian models with MCMC and municipality and time-varying parameters, achieving **0.878 out-of-sample R^2** while quantifying uncertainty and changing spatiotemporal relationships.
- First author of a paper accepted to **IEEE IGARSS 2026** on spatiotemporal dengue forecasting; journal manuscript on spatiotemporal dengue modeling and Bayesian inference in preparation.

Graph Theory and Combinatorics Research

Sep 2024 - Dec 2024

- Co-developed **Alytiqcon**, an interactive Python/Pygame graph-analysis application supporting construction and analysis of directed, undirected, weighted, and unweighted graphs.
- Implemented **Fleury's and Hierholzer's algorithms**, backtracking, and dynamic programming for graph traversal, pathfinding, and combinatorial search; **deployed the application to the browser** with Pygbag.
- Co-authored research on graph algorithms, combinatorial optimization, and real-world routing applications; invited to present at **TUMC 2024**.

M3 International Mathematics Competition

Feb 2025

- Earned **Honorable Mention (top 20 of 800 submissions)** in the M3 Challenge; co-authored a 30-page modeling paper on heatwave risk, power demand, and infrastructure vulnerability in Memphis.
- Built Python models for indoor heat transfer, 50-year electricity-demand forecasting, and neighborhood vulnerability, including a neural network trained over **5,000 gradient-descent iterations** and sensitivity analysis.

LEADERSHIP & ACTIVITIES

MIT Native American and Indigenous Association | President

Aug 2026 - Present

- Lead chapter programming, conference participation, industry outreach, and professional-development initiatives for MIT students.

AISES, MIT Chapter | Executive

Aug 2026 - Present